Analysis of optical timestamps using the PPS flash device

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1pps Flasher Method:

The 1pps flasher uses a GPS receiver to run a program on an Arduino Nano that causes an LED to light up for a specific duration when the 1pps signal begins.

A faint flash in the optical path slightly brightens the image in dark areas. If the LED is bright enough, this brightening of the image background can be measured in a video. Analyses, such as photometry, are hardly affected - if at all - by this. In any case, during star occultations, the object is not so heavily illuminated that it affects time measurements.

The following section provides a step-by-step description of how the flashes from the 1pps flasher can be used to determine time measurements for scientific purposes.

Timestamp Accuracy:

For lack of a better option, an existing DVTI camera (1) was used and operated in a mode with a 2-millisecond exposure time, corresponding to 500 frames per second. Videos were created with analyzable optical timestamps from the 1pps flasher and compared to the start and end times of each frame from the DVTI camera using the time stamps displayed at the bottom of the image. Of particular interest here are the start of the frame and the start of the flash.

By comparing the results - using a “white tag” from PyMovie - with the information in the image, it was determined that the optical timestamp yielded times for the start of a second (ss,000) that were exactly up to 1.5 ms earlier.

Certainly, more accurate measurements can be taken with better equipment, such as EXTA (2). However, for the purpose of star occultations, the available results appear to be sufficient to provide scientifically accurate data and reports using the evaluation method described here.

Recording a Video with the 1pps Flasher:

For scientifically valid time measurement, it is very important to take into account the duration of the flash and how this affects the video camera being used!

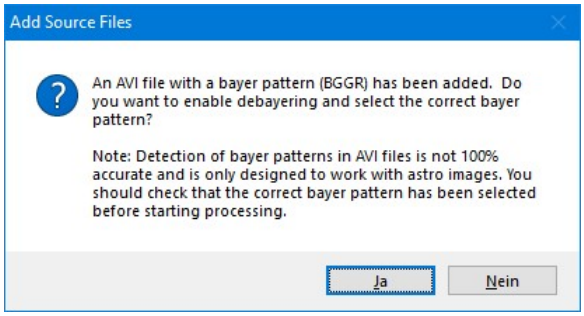
The author uses the 1pps flasher with the Seestar S50. Therefore, everything in this article refers to this specific application. If the flasher is used with other cameras, the user must conduct their own measurement series and tests using the methods and principles described here in order to meet scientific standards.

Analysis Using PyMovie / PyOTE (3):

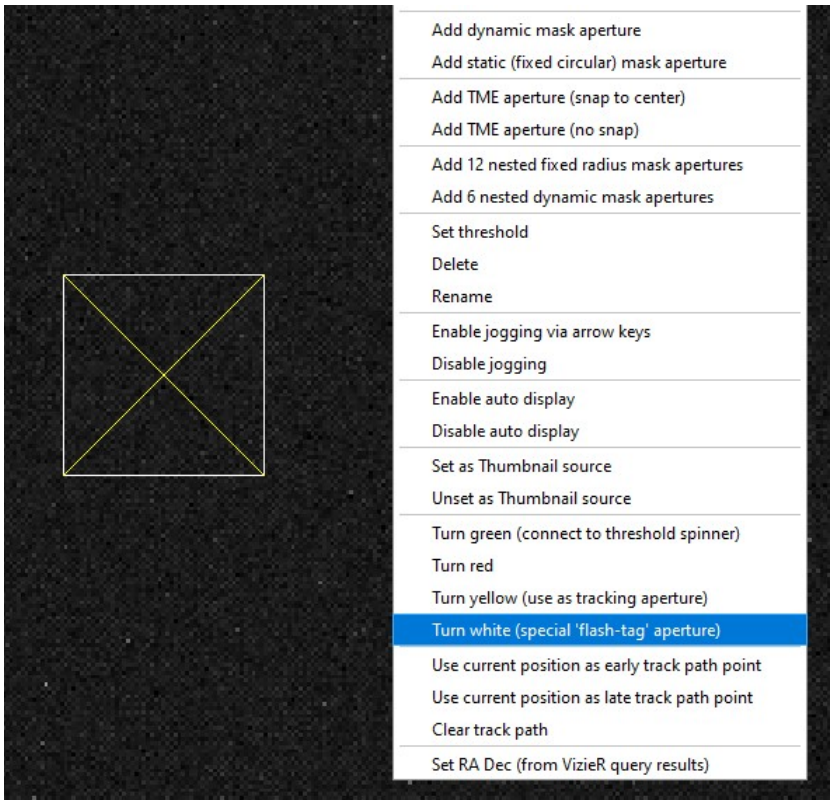
White tag in PyMovie:

Once PyMovie and PyOTE are installed and running smoothly, open the video in PyMovie and define the measurement fields. The user should already be familiar with this procedure.

Important Note: A video recorded with the Seestar S50 must be captured with “Raw enabled.” This results in a very large AVI file. Due to the proprietary Bayer pattern (BGGR), this file must first be converted and vertically flipped. The author accomplishes this using PIPP (4). The configuration file S50_debayer_flipV.pof is loaded via “load options” with all necessary settings and is included at the end of this article. By clicking “do processing” and “start processing,” PIPP creates an AVI file that is used for further processing.



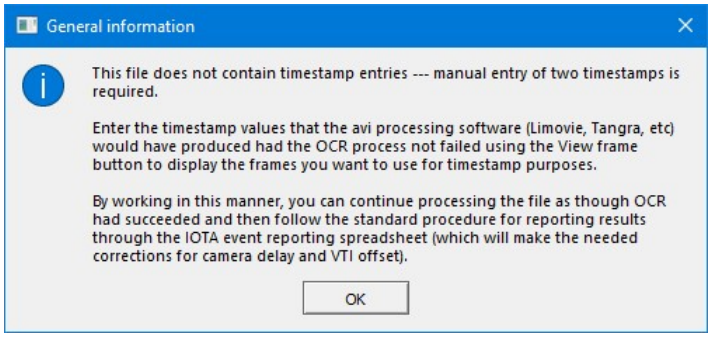
In PyMovie, right-click and select “Add dynamic mask aperture” to add a mask in a dark area without bright spots (stars); then, in the second step, use “Turn white (Special ‘flash-tag’ aperture)” to change it into a white-bordered measurement field.

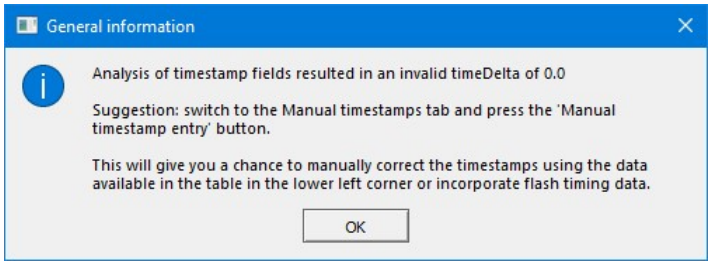


After analyzing the video with PyMovie, the saved light curve (*.csv) is opened with PyOTE.

Light curve of a Seestar S50 in PyOTE:

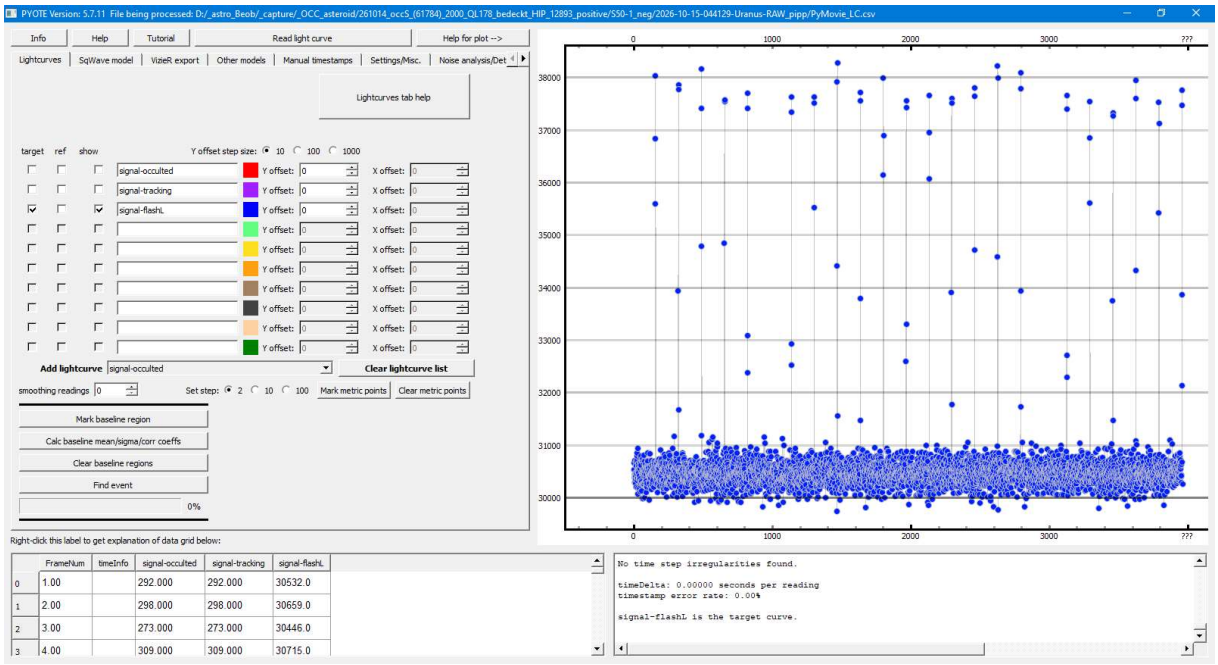
When opening a light curve from the Seestar S50, two messages immediately appear indicating that there are no timestamps, which is already known.





Later, we'll show how to define these timestamps for PyOTE.

But first, in the “Lightcurves” tab, we'll make only the white-tag signal visible.



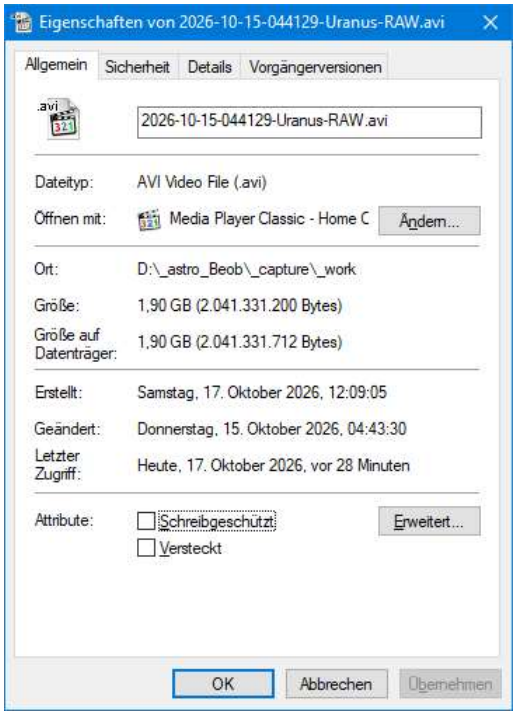
Here you can see the light curve of the faint flashes as peaks. The 1pps flasher was set to mode #85, in which a flash occurs every 5 seconds but the 55th second is skipped. As a result, the start of each minute is easily recognizable by a noticeable gap. The peak following a gap is therefore assigned the time ss.000 seconds.

The video is almost exactly 2 minutes long because the author used the MacroDroid (5) robot app.

Note: This app makes it possible to automatically record short videos simultaneously using multiple S50s at remote locations.

Classification by the exact minute and hour:

In Windows, the properties of this video look like this (Alt+Return):



There, under “Geändert,” you can see that the video ended at 04:43:30, provided the file has not been modified since then. Note: Be sure to back up your data!

This time was in Central European Summer Time (CEST), so the end time should be 02:43:30 UT, and the start time is 2 minutes earlier. However, this is only accurate to the second, since it’s based on the smartphone’s time, which is usually off by a few seconds from the actual time. The reader will now understand why the minute markers in the video are so important!

The attentive reader will have noticed that the peaks have multiple points - sometimes 3 and sometimes 4. This is due to the exposure time used here of 0.030 seconds with a flash duration of 0.080 seconds.

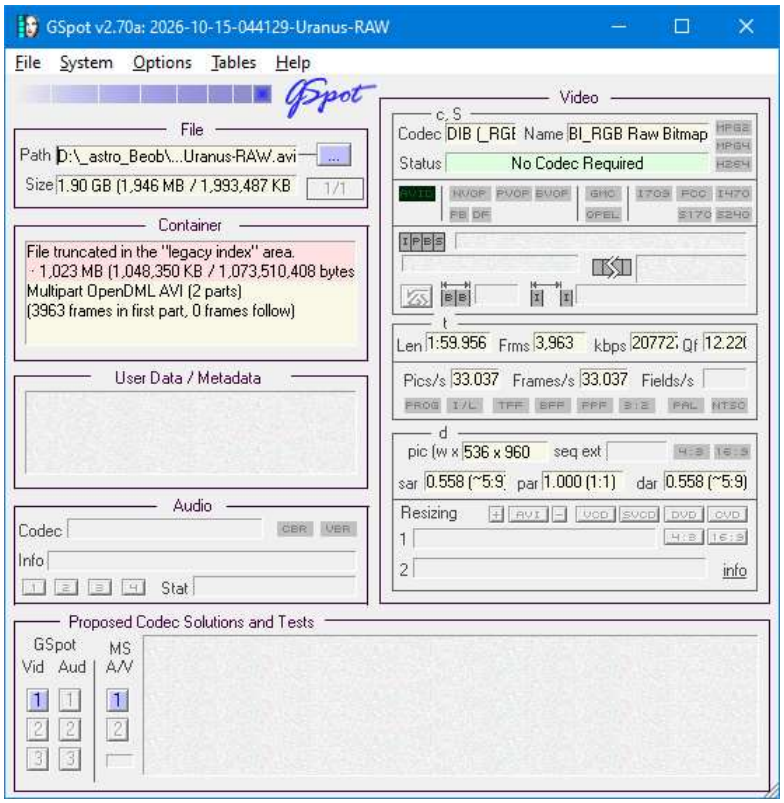
Note: Initially, the code (6) for the 1pps flasher was set to 40 milliseconds; the author now recommends 40 milliseconds (ONdurNorm = 80), because otherwise, with an exposure time of less than 0.08 seconds in 1X mode (full format of the S50), flashes may “disappear,” since only about 12.5 frames per second can be achieved in this mode (7). This makes it impossible to determine the minute. Tip: If this does happen, you can watch the mp4 video and, with a bit of luck, still spot the gaps in the minute markers there.

The following section therefore describes a light curve with an 80-millisecond flash. This is important for accurate time markers and understanding. Every reader using a 1pps flasher with a 40-millisecond flash must take this into account. How this is done will be shown later.

You can easily determine how long the device’s flash lasts even during the day: Operate the S50 in 4X mode with “closed arm,” set the exposure time to 0.01 seconds, record a short video with 1pps flashes at maximum intensity, and analyze the light curve. 1pps flashers with 40 ms show peaks with 4 or 5 points, whereas those with 80 ms show 8 to 9 points.

So it’s not only important to know what exposure time you used while filming, but also how many frames per second the camera captured! Note: You shouldn’t push the camera to its limits, because a camera that’s otherwise time-constant may suddenly start skipping frames, and the start times of the images may also show uneven intervals!

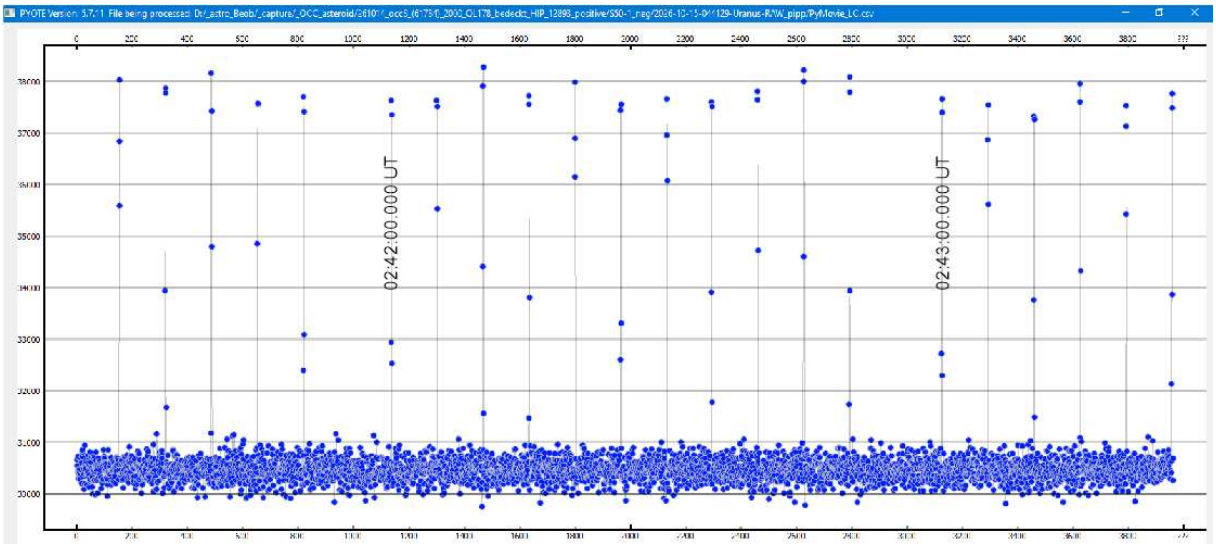
The author quickly determines the frames per second using the simple, old program GSpot (8)—“Pics/s” here shows 33.037 fps.



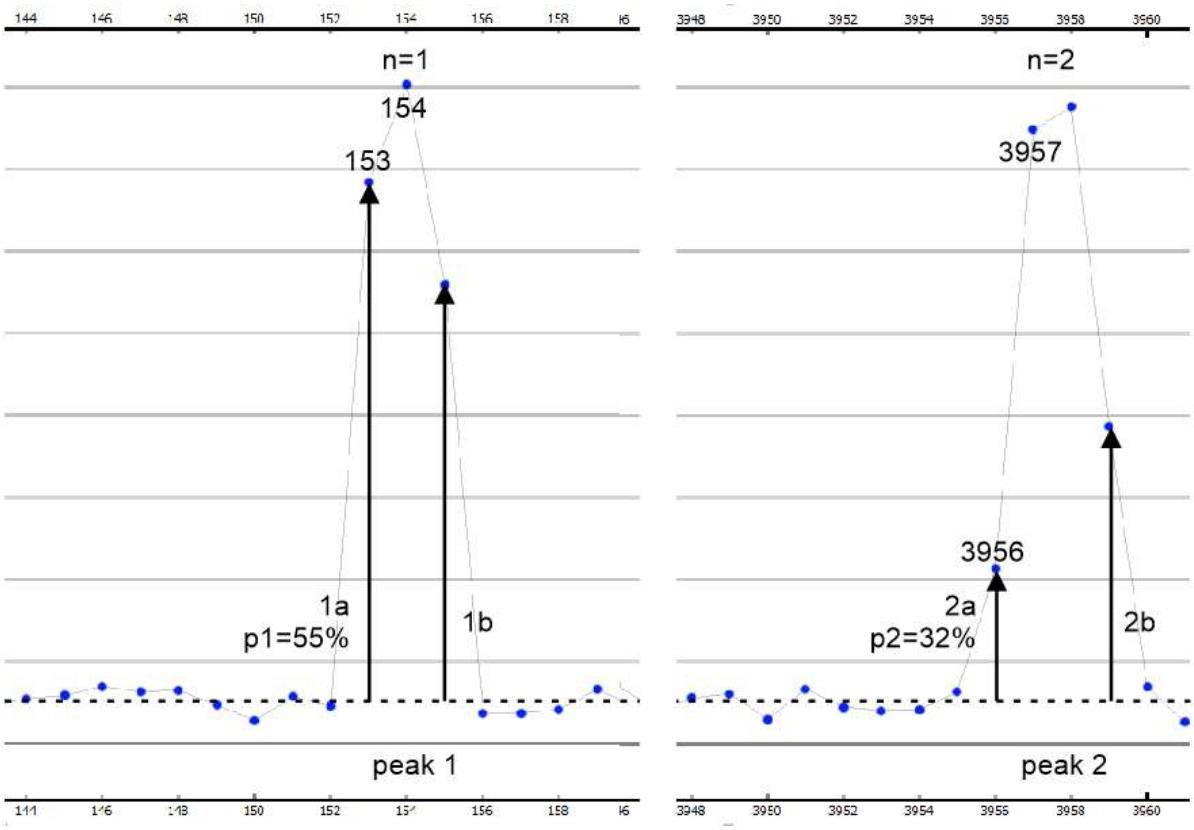
1. **Case: Exposure time shorter than the flash duration:**

Back to the peaks with a 80-ms flash duration and a 30-ms exposure time. This means a flash spans 2.66 frames. So we can expect to see 4 points every time: two frames with full exposure, and a split flash before and after them. However, it can also happen that only three frames are illuminated by the flash.

The following image shows the entire light curve of the flashes with the time-stamped minute starts that have already been determined..



Here, for example, the first and last flashes are displayed and measured so they can be used later in PyOTE:



Only the first percentage portion of the split flash is needed. This is calculated as follows:

$$p = \frac{a}{a+b} \qquad \text{where p1=55\% and p2=32\%}$$

It is noticeable that the lengths of a and b do not even approximately correspond to the height of the points at full exposure, but that is actually not important. Tests have shown that the dimensions and calculations shown here yield better results.

How to measure the length of the arrows is up to each reader, as there are many ways to do so. The accuracy is also sufficient as long as it is better than the noise of the baseline, whose average value can be estimated by eye in the vicinity of the peak.

For clarity: Image 154 or 3957 is the first image with full exposure. It is given the timestamp 02:41:30.xxx or 02:43:25.yyy

xxx/yyy is the proportion of the flash in the previous image, 153/3956.

The ratio p1/p2 is applied to the time remaining after subtracting the fully exposed images.

Coincidentally, the first peak (left) has three points and the last (right) has four. The following table illustrates the result.

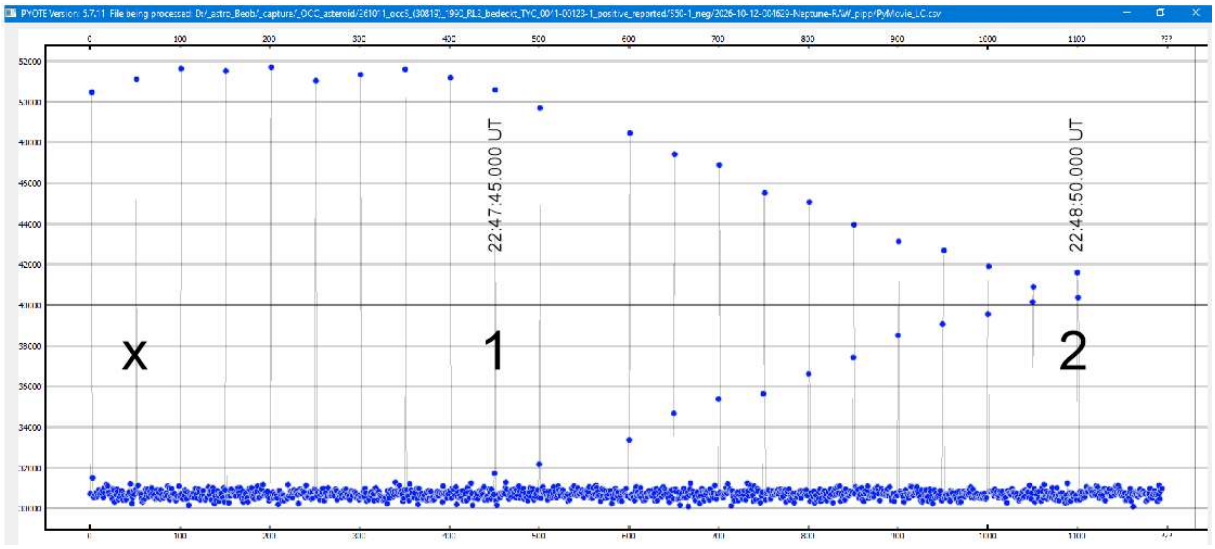
t_{flash} = 80ms	n at t_{exp} = 30ms	Remaining time	p	Frame	Partial flash	**Frame**	**Timestamp**
left	1	50ms	55%	153	27.5ms	**154**	**02:41:30,044**
right	2	20ms	32%	3956	6.4ms	**3657**	**02:43:25,026**

The following equation covers both cases.

$$t_{exp} < t_{flash}: \qquad xxx = p * (t_{flash} - n * t_{exp}) \qquad Format\ \ hh{:}mm{:}ss.xxx$$

2. Case: Exposure time longer than flash duration:

A light curve like this might look as follows.



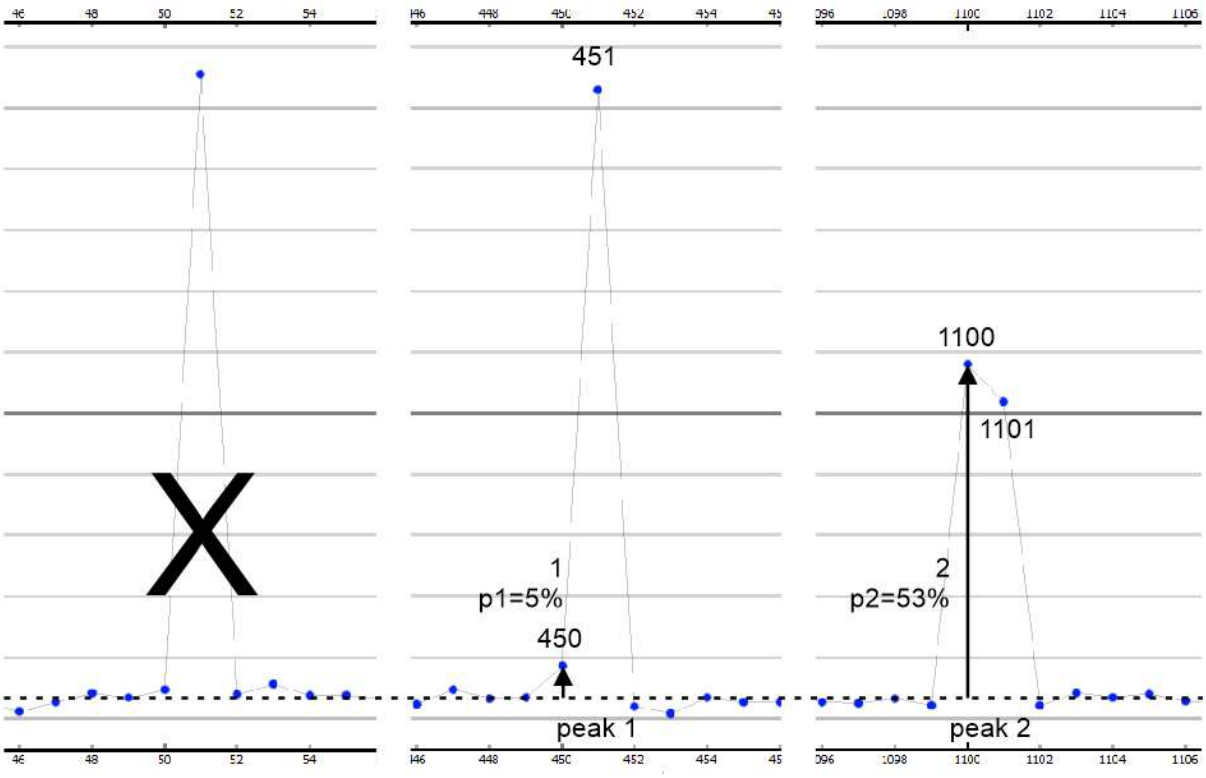
In this case, a flash fits entirely within a single frame duration, as seen with the peaks on the left side of the image. Only a single dot is visible there. Such a peak is unsuitable for analysis because the difference between the flash duration and the exposure time further exacerbates the inaccuracy in the time measurement.

If, for any reason, it becomes necessary to use a flash as in X, the center of the image should be used as the reference point for the flash, since the flash occurred somewhere between the beginning and the end. An additional uncertainty equal to half the difference between the exposure time and the flash duration should be taken into account. This must be added to the uncertainty associated with the analysis of a stellar occultation.

$$t_{exp} > t_{flash}\ for\ 1\ frame: \quad \Delta t_{1frame} = (t_{exp} - t_{flash})/2$$

only 1 frame	Exposure time t_{exp}	Flash duration t_{flash}	Difference	**Frame**	**Timestamp**	**<u>Additional</u> uncertainty Δt_{1frame}**
X	100ms	80ms	20ms	**51**	**22:47:05.050**	**±0.010**

If the event occurred in the left part of the image and you have to use peaks to the right of it, as in this case, you should try to use peaks that are as far apart as possible, such as those labeled 1 and 2.



Now, using peaks 1 and 2, interpolation can be performed much more accurately than with X:

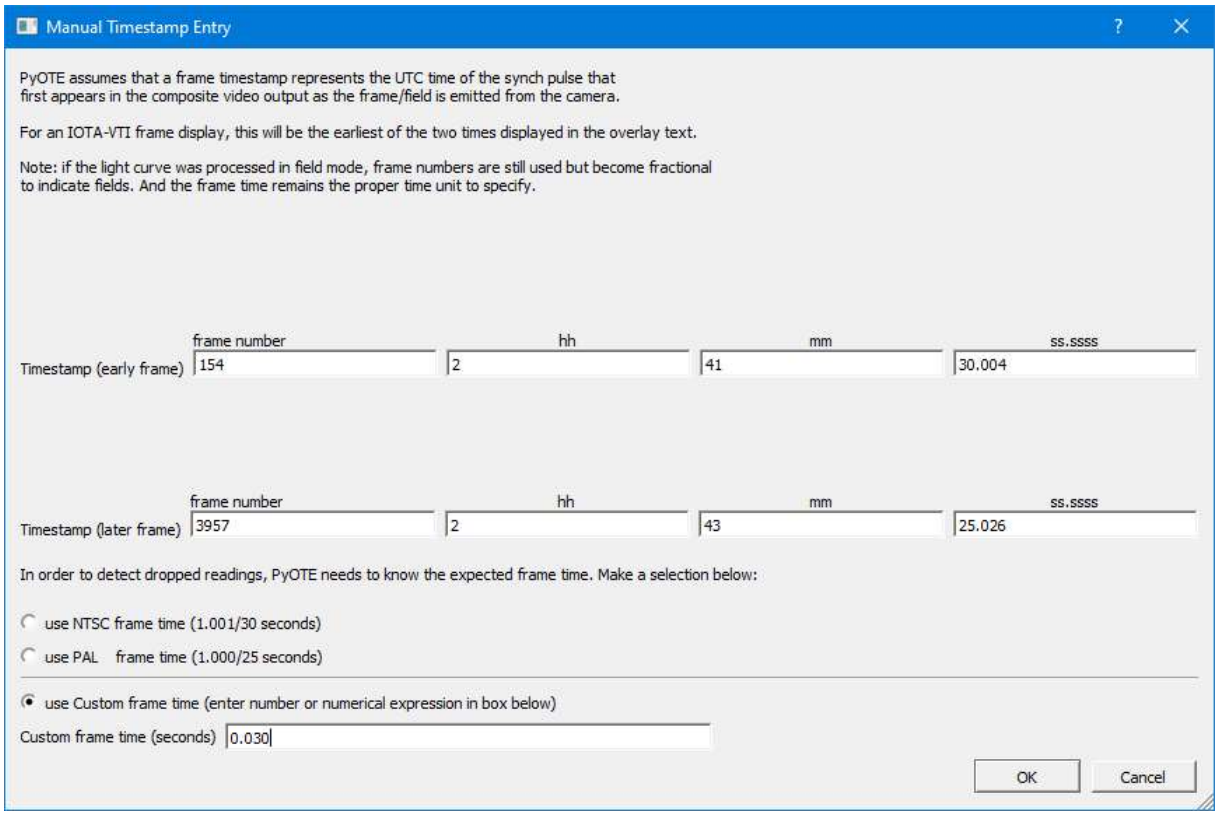
t_{exp} = 100ms	Flash duration	p	Frame	Proportion	**Frame**	**Timestamp**
1	80ms	5%	450	4.0ms	**451**	**22:47:45.004**
2	80ms	53%	1100	42.4ms	**1101**	**22:48:50.042**

The following equation covers both cases.

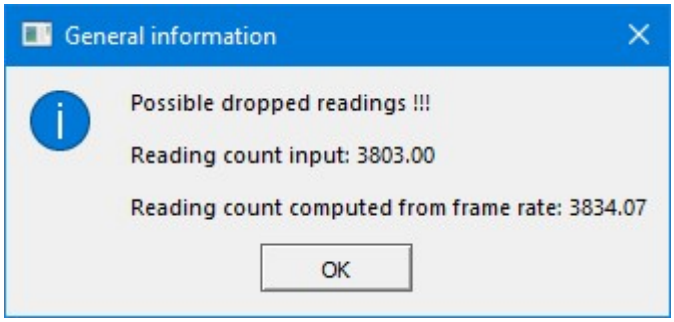
$$\boldsymbol{t_{exp} > t_{flash}}: \qquad \boldsymbol{zzz = p * t_{flash}} \qquad \boldsymbol{Format\ \ hh{:}mm{:}zzz}$$

Definition of timestamps in PyOTE:

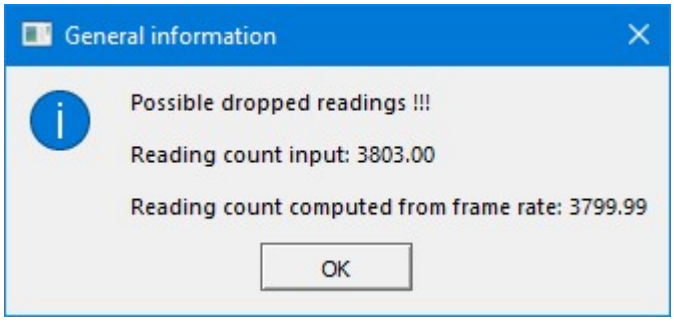
Under the “Manual timestamps” tab and the “Manual timestamp entry” button, enter the times in the window that opens (values for the first case shown above).



The interpolated values are entered.



A deviation of 31 frames would correspond to nearly one second, but apparently, instead of the 0.03-second exposure time indicated by GSpot (above), the S50 took 0.030269 seconds at 33.037 fps. This would result in a deviation of only 3.0 frames.

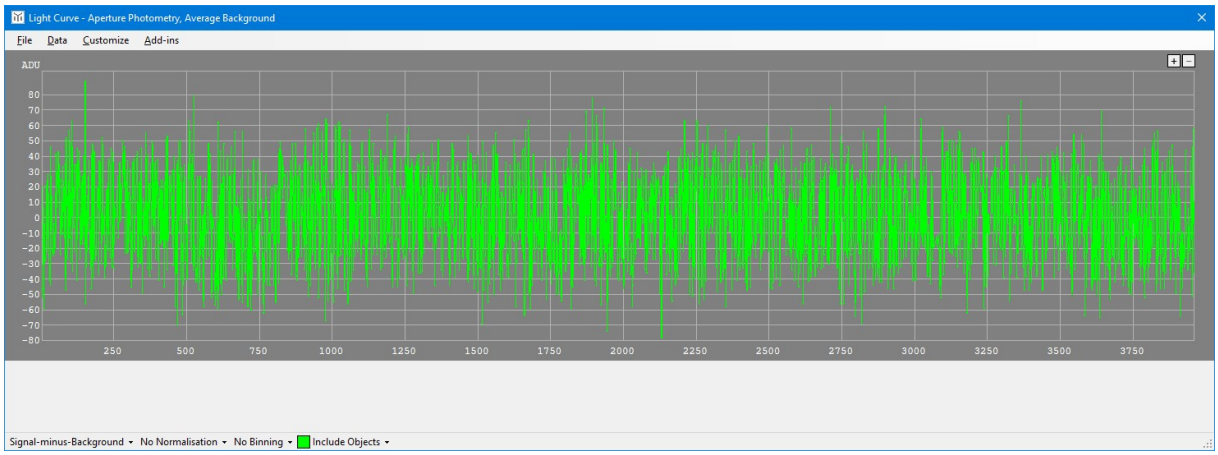


That would be acceptable. However, in the vicinity of an observed occultation, one should check the 5-second peaks for deviations and, to obtain more precise time measurements, analyze nearby peaks using the method described and enter them as timestamps in PyOTE. This will help identify any deviations.

Analysis using Tangra / AOTA:

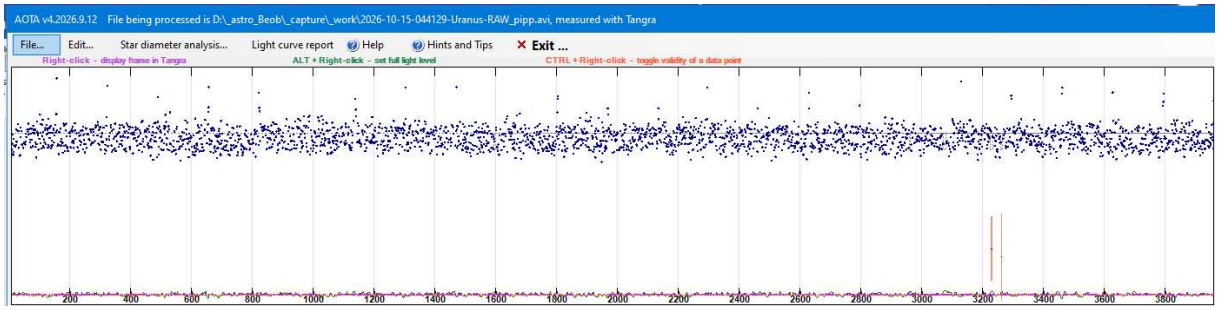
The author is inexperienced with this procedure.

You should set a measurement point at a location without stars.
After analyzing with Tangra, continue without times—“continue with no times.”



However, it is clear that the peaks are barely visible. When analyzing the data with Tangra, you would need to set the flash to be significantly brighter.

Then, in the AOTA settings, set “binning: 1 frame” and “running average: 1 bins.”



You can see how difficult it is to detect the peaks even in AOTA. Reliable detection cannot be achieved with this weak flash, whereas PyOTE (see above) had no problem with it at all.

Unfortunately, the author cannot provide any further insights at this time.

Translated with DeepL.com (free version)

References

1 Scientific CMOS Video Camera With Digital Time Insertion
https://dvticam.com/

2 EXposure Time Analyzer - EXTA
http://www.dangl.at/exta/exta_e.htm

3 IOTA: Installation and use of PyMovie / PyOTE

https://occultations.org/observing/software/pymovie/
PyMovie installation
https://pypi.org/project/pymovie/
PyOTE installation
https://pypi.org/project/pyote/

4 PIPP v2.5.9
https://web.archive.org/web/20230531163522/https://sites.google.com/site/astropipp/downloads

5 MacroDroid app for Android
macrodroid-automatisierung.de.appcompare.net

6 source code: 1pps_flasher
https://github.com/leontodon1/1pps_flasher

7 Folie 6, own presentation of the author
28. Kleinplanetentagung, VEGA-Sternwarte Haus der Natur, 19.-21.Juni, 2026
http://ftp-europlanet.de/wp-content/uploads/2026/06/Vortrag_BjKa_KP28_2026.pdf

8 GSpot
https://www.videohelp.com/software/GSpot

S50_debayer_flipV.pof

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[PIPP OPTIONS FILE]
BATCH_MODE True
JOIN_MODE False
PLANETARY_MODE False
FULLDISC_MODE False
CLOSEUP_MODE False
PLANETARY_ANNIMATION_AVI_MODE False
PLANETARY_ANNIMATION_GIF_MODE False
ISS_MODE False
AVI_ARCHIVE_MODE False
FRAME_LIMIT False
FRAME_RANGE_TYPE 0
START_FRAME_VALUE 1
END_FRAME_VALUE 1200
NUM_FRAMES_VALUE 1200
DARK_FRAME_SUBTRACT True
SAVE_MASTER_DARK False
REPLACE_HOT_PIXELS False
HOT_PIXEL_THRESHOLD 90
DARK_DATA_HAS_BAYER False
FLAT_FRAME_CALIBRATE True
SAVE_MASTER_FLAT False
FLAT_DATA_HAS_BAYER False
FLAT_DARK_FRAME_SUBTRACT True
SAVE_MASTER_FLAT_DARK False
BINNING 0
BIN_CALC 0
DEBAYER_ENABLED True
PROTECT_BAYER False
DEBAYER_TYPE 0
DEBAYER_PATTERN 1
RAW_HPF False
RAW_DEBAYER True
RAW_DEBAYER_ALGORITHM 0
RAW_COLOUR_SPACE 0
RAW_HIGHLIGHTS 1
TRUE_Y800 False
SER_BIT_DEPTH 0
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INPUT_FRAME_COLOUR 0
IP_FRAME_DROP_VALUE 0
AVI_OLD_FORMAT False
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AVI_OLD_FORMAT_MAX_SIZE 0
AVI_CODEC 0
AVI_CODEC_QUALITY 10
DATE_TIME_EXTRACT False
WJUPOS_CONV_UT True
WJUPOS_DATETIME_FORMAT 0
WJUPOS_DATETIME_TEXT YYYYMMDD_hhmmss
WJUPOS_INPUT_FPS 60
INPUT_GAIN 1
GAMMA 1
MEDIAN_FILTER False
CHECK_PLANET False
MIN_PLANET_SIZE 15
EDGE_IN_SHADOW 0
CENTRE_PLANET False
AUTO_THRESHOLD True
THRESHOLD 40
OVEREXPOSED False
OVEREXPOSED_VALUE 20
ENABLE_CROP False
CROP_WIDTH 448
CROP_HEIGHT 448
X_OFFSET 0
Y_OFFSET 0
HISTO_STRETCH False
HISTO_STRETCH_VALUE 75
HISTO_STRETCH_NO_BP False
RGB_STRETCH True
KEEP_IMAGE_FILENAMES True
NO_FRAME_RENUMBER False
ADD_LEADING_ZEROS True
ADD_QUALITY_VALUE False
STABILISE_NONE_MODE True
STABILISE_OBJECT_MODE False
STABILISE_SURFACE_MODE False
SURFACE_FEATURE_TRACKING_EN False
STABILISE_EN False
AFB_X 150
AFB_Y 150
AFB_WIDTH 128
AFB_HEIGHT 128
AFB_REJECT True
AOI_ENABLE False
AOI_X 10
AOI_Y 10
AOI_WIDTH 300
AOI_HEIGHT 300
AOI_REJECT True
AOI_CROP True
CONV_MONO True
CONV_MONO_TYPE 0
FLIP_X False
FLIP_Y True
ROTATE 0
RESIZE_FRAMES False
RESIZE_KEEP_ASPECT_RATIO True
RESIZE_WIDTH 150
RESIZE_HEIGHT 150
ANIM_FWD_ENABLE True
ANIM_REPEAT_FWD_LAST_ENABLE False
ANIM_REPEAT_FWD_LAST_VALUE 1
ANIM_REV_ENABLE False
ANIM_FWD_ENABLE True
ANIM_REPEAT_REV_LAST_ENABLE False
ANIM_REPEAT_REV_LAST_VALUE 1
ANIM_REPEAT_ALL_ENABLE False
ANIM_REPEAT_ALL_VALUE 1
ANIM_REPEAT_REV_DISABLED_LAST True
ENABLE_QUALITY False
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QUALITY_LIMITING True
QUALITY__LIMITING_TYPE 0
QUALITY_FRAMES 1200
QUALITY_REORDER True
DEFAULT_QUALITY True
NINOX_QUALITY False
HISTO_QUALITY False
TOTALPIXEL_QUALITY False
QMIN 3
QMAX 5
QINC 1
ENABLE_QWEIGHT False
CREATE_OUTPUT_SUBDIR False
SUBDIR_NAME pipp_
SUBDIR_APPEND_TIME True
OUTPUT_DIRECTORY Default (Source File Directory)
OUTPUT_PREFIX
OUTPUT_SUFFIX _pipp
OUTPUT_AVI True
OUTPUT_SER False
OUTPUT_BMP False
OUTPUT_TIF False
OUTPUT_PNG False
OUTPUT_FITS False
OUTPUT_GIF False
AVI_FPS_AS_INPUT True
AVI_FPS_VALUE 60
GIF_REPEAT_ALL_FOREVER_VALUE True
GIF_FPS_AS_INPUT False
GIF_FPS_VALUE 3
GIF_COLOUR_QUANT_TYPE 0
GIF_ADV_OPTIONS 1
GIF_UNCHANGED_BORDER_EN True
GIF_UNCHANGED_BORDER_TOL 5
GIF_TRANS_PIXEL_EN True
GIF_TRANS_PIXEL_TOL 5
GIF_BIT_DEPTH_EN False
GIF_BIT_DEPTH_VAL 7
GIF_LOSSY_COMP_EN False
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RGB_SPLIT False
FRAME_DROP_VALUE 1
WJUPOS_NAME_ENABLE False
WJUPOS_TIME 0
SAVE_REJECTED False
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